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From COVID to conflict: Understanding the deriving forces of environment and implications for natural resources

Qasim Raza Khan [a,*], Mao Xinshu [b,**], Ghulam Muhammad Qamri [c], Ahmad Nawaz [d]

[a] School of Economics, Beijing Technology and Business University, Beijing, PR China
[b] School of Business, Beijing Technology and Business University, Beijing, PR China
[c] School of Economics, Nankai University Tianjin, 300071, PR China
[d] Department of Economics, University of Sahiwal, Sahiwal, Pakistan

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ABSTRACT

In the contemporary world, the importance of natural resources is increasing day by day especially due to extraordinary circumstances, i.e., COVID-19 and global conflicts. The abundance of natural resource is considered competitive advantage and crucial for sustainable development. However, the role of natural resources can be questionable especially if its impact on the economy is negative. Sustainable use of natural resources is currently the biggest challenge for governance. Following these footprints, the study aims to revisit a novel perspective of natural resources in the context of global conflicts using data from Asian economies for the period of 1996–2020. In this pursuit, this study investigates how governance balances macroeconomic variables with sustainable development to account for effective climate change adaptation, mitigation efforts and integral to control conflicts. The second-generation test of CIPS and CADF are used to deal with cross-sectional dependence issues and Westerlund cointegration to estimate long-run relationships. Furthermore, the long-run coefficients are estimated by the PMG estimator using dynamic panel ARDL approach. The findings confirm that surpassing the threshold level of governance is essential to promote environmental quality and preservation of natural resources. The region needs to promote steward policy for resources. This can take the form of nationalizing resource assets, increasing taxes and royalties on resource extraction to ensure sustainable development. The handlers need to design polices supportive to renewable energy consumption, endorse IT based industry solution, encourage high-tech inward FDI, promote green financing and support sustainable development.

1. Introduction

Natural resources reserves could be either a curse or blessings for economies around the globe; however, the role of global pandemics, i.e., COVID-19 and the importance of global conflicts, i.e., Russia-Ukraine war become important to coup with its negative impact and to channelize its positive impact on the economy. Similarly, economies with abundance of natural resources, i.e., Asian economies then become the center of discussion and gets high geopolitical importance in the global arena. To accelerate economic growth emerging economies, improve governance mechanisms, endorse technological innovation, develop financing mechanisms, promote natural resources stewardship, and encourage inflows of foreign direct investment to meet the funding requirement for the development of both operational and economic activities (Umar et al., 2020). Despite its significance, the inflow of such funds engenders certain costs in the form of natural resources utilization and environmental degradation like developments in financial system spur the economic activities which posit negative effects on environment. However, the progress in technology, stewardship in natural resources, and governance can minimize these effects over the time. Where every country is in a dire need to increase the economic prosperity, but the real challenge confronting is about how to ensure growth without deteriorating resources and environment. Sustainable development is the goal of every nation in current era. Every country wants to design governance system which manage the natural resources and attracts foreign direct investment to increase the inflow of funds to boost

* Corresponding author.
** Corresponding author.
E-mail addresses: qasi.raza@gmail.com (Q.R. Khan), maoxinshu@163.com (M. Xinshu), gmqammar@hotmail.com (G.M. Qamri), ahmadnawaz@uosahiwal.edu.pk (A. Nawaz).

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economic system, enhancement of technology to support economy infrastructure with modern techniques to make it more efficient and effective kind of financial system to fulfill financial needs of a country for development. But the question arises on governance system; if economic growth achieves at the price of overutilization of natural resources and environmental degradation, it means governance is just a documented goal? Does it mean governance didn't support sustainable development? According to current statistics, USA and China are the biggest emitter of carbon (CO_2), almost 44% emissions by such countries claim the best governance system of the world. If the business as usual continues, the global extraction of natural resources reaches 110% in 2060. Does it mean that governance has contradictory relation with economic growth and environment? Sustainable use of natural resources is the biggest challenge for governance now? Literature claims, best governance system spur the economic activity by attracting high foreign direct investment, develop modern and advanced infrastructure, promote stewardship of natural resources, supportive to efficient and effective financial mechanism. This study expects to unwrap the impact of macroeconomic variables on the most emerging set of countries. This region has abundance of natural resources and top carbon emitter around 60%, so it's quite important to study these variables in this perspective of this region.

It is an evident that both developed and developing countries are facing the issue to attain balance between consumption of natural resources and economic growth. The deterioration in environment is the cause of adverse conventional consumption of NR and greenhouse gas emission. In the same pursuit world also developed some understanding to deal with the issues for upcoming generation. Like, convention held by United Nation on climate change countries agreed to keep temperature below than 1.5 °C by 2020 (Nathaniel and Bekun, 2020). Same like as per the United Nation Development Programme (2016), the signatory countries agreed on the preservation and use of natural resources while promoting the culture of NR stewardship. So, the future generation can take their own share (Sinha and Sengupta, 2019). Latest reports of United Nations on sustainable development revealed that the developing nations are performing below the lines in environmental indicators such as green growth, technology supportive solutions, advanced financial mechanism, ineffective institutional role and in many others ecological parameters (UNESCAP, 2021). These concerns were also highlighted by United Nations during the appraisal of ecological SDG's (UNESCAP, 2020). It pointed out that Asian economies are laggard quite behind to achieve the climate action SDG 12 & 13. In poverty baseline report of 2019, World Bank highlighted less pro ecological environmental and sustainable resources policies are the biggest hindrance for development in Asia (Fallesen et al., 2019).

Asian countries contain a significant portion of the world's natural resources. Countries in this region have the world's largest oil and natural gas reserves, with countries like Saudi Arabia, Iran, Iraq, and Kuwait possessing large reserves of these resources. In addition, China is the world's largest producer and consumer of coal, and it also possesses significant reserves of rare earth minerals, which are used in the production of many high-tech products. Other Asian countries are known for their reserves of specific resources. For example, Indonesia is a major producer of nickel and tin, while Mongolia is known for its large reserves of copper, gold, and coal. Kazakhstan is rich in minerals like uranium, zinc, and lead (Shittu et al., 2021). It's worth noting that the development and exploitation of natural resources can have significant economic, environmental, and social impacts, and it's important for countries to balance the benefits of resource extraction with the potential costs and risks.

Comparatively, Asia stands as one of the most dynamic economic regions and also the largest CO_2 emitter in the world. Emerging and developing Asia consists of 48 countries with a population of more than 4.6 billion (60% of the world's population), which in total represent a nominal gross domestic product (GDP) of about $3.58 trillion; $65.44 trillion in GDP purchasing power parity and 5.7% annual GDP growth rate (WDI, 2020). Following the lead, the largest carbon dioxide (CO_2) emissions in the world is also Asia Pacific i.e. currently the world emissions is 35 billion tones every year, out of which around 20 billion (56%) world emission is from Asian countries (WDI, 2020). China alone produced 28.8 percent of global CO_2 emissions (Statista, 2020). Globally, CO_2 emissions trend has been rapidly increasing over the time from 1968 till 2018, approximately by 23.7 billion metric tons. Moreover in 1990, Asia emission was almost 6.6 billion tons that subsequently increased to 20 billion tons in 2020, which is almost 3 times more than before. If we talk about top 10 emitters of the world, 8 countries i.e. (China, Russia, India, Iran, South Korea, Japan, Indonesia and Saudi Arabia) are from Asia region. The alarming figures warrant a dire need for Asian countries to shift towards low carbon emissions in future. Literature recommended that this region needs technological supportive solutions, foreign direct investment with modernized infrastructure, environment friendly policies, green finance-based solution in financial policies, sustainable management of natural resources, transformation of traditional pollution intensive industry solutions into highly ecological and green development supportive information centers to uplift the economic growth (Destek and Sarkodie, 2019).

This study contributes to existing literature in numerous ways: (i), It will test association in the context of Asian countries on available longest data set (1996–2020). As our best, this is first kind of study which analyses this nexus on all Asian countries, before this most of the studies used regional sub-sets data like East- Asia; south Asia etc. (ii), Sustainable use of natural resources and ecological development is biggest challenge for governance at present, existing literature will seek answer to it (iii), In many of the past studies different methods have been used for improving institutional quality, like the rule of law, voice and accountability (Jadhav, 2012); quality governance (Ferreira and Matos, 2008), corruption (Javorcik and Wei, 2009); and political unpredictability (Herrera-Echeverri et al., 2014). This study used all the six indicators of governance developed by WGI. (iv), The study provides a more in-depth portrayal of the interrelationship among climate change and NR conflict. Finally, the study provides an in-depth depiction of CO_2 emissions and inward foreign direct investment relationship with the perspective of natural resources.

2. Literature Review

The relationship between climate change and conflict is complex. Climate extremes and variability are threatening the livelihoods, food security, health and wellbeing of millions of people worldwide, with those living in fragile and conflict-affected states disproportionately affected. Conservation and preservation of natural resources is very important for the sustainable development and survival of human life. A nation with the diverse and abundance of natural resources can grow more than the fewer one. Especially, the socio-economic and geopolitical change made the world uncertain and debatable (Sinha and Sengupta, 2019). The continuous exploitation is not just reducing NR availability in future, it also affecting ecosystem and environment unpleasant and harmful. For healthy communities, there is sturdy need to provide clean air, revitalized natural resources, and non-toxic environmental solutions. Sustainable development creates a balance between economic prosperity and natural resources consumption. The prime issue regarding environmental protection and sustainability revolves around how well natural system functions can works and equilibrates ecological needs. Currently, the world is confronting with observable differences on the balance amidst economic development and environmental degradation (Saint Akadiri et al., 2019). In recent years, growing energy depletion from fossil fuels are one of the main reasons of pollutant environment (Huang et al., 2020; Anwar et al., 2021). This rapid degradation acquired the intention to deal with such environmental deterioration (Li et al., 2022). Climate change has attracted the attention of policy maker, researcher, academicians, and the institutions due to rapid climate change over the last 2 decades. This massive

environmental change is very warm and unhealthy for life, ultimately occurs because of excessive energy pollution, industrial explorations, fuel intensive consumption, weak intuitional role, less ecological and environmental polices (Ulucak, 2020; Alzard et al., 2019; Jahanger, A. et al., 2022).

2.1. Nexus between natural resource and CO_2 emissions

Natural resources are materials or output that people discover and use to support economic activities or meet human needs. The availability of natural resource has a major role in any country to gain an economic growth or power in modern era. The richness of natural resources is considered a competitive advantage for any country in economic development. The overexploitation and exploration of natural resource is not just decreasing production efficiency also deteriorating the environment. Sustainable use of natural resources and green growth is crucial because it assist to avoid unnecessary use of resources without affecting future generation. The consumption of natural resources in countries has increased dramatically over the past several years and this abundant utilization to eventual level put an adverse impact on the environment. The conventional method of high consumption in non-renewable energy resources has become the reason of environmental deterioration (Zhang et al., 2018; Chen et al., 2022). In some studies, natural resources volatility is considered supportive to pollution reduction by eliminating the conventional utilization of energy resources (Balsalobre-Lorente et al., 2018).

The adverse impact of natural resources consumption on environment can be eliminated by improving institutional quality, advanced industrial solutions, high-tech inward FDI, and country governance system. Some studies claim that democracy improves environmental quality by promoting voice, freedom of expression, and public understanding of how to effectively manage natural resources (Barbier, 2015; Wang et al., 2019). The undermining of environmental governance and resources depletion are the cause of ineffective and poor implication of government policies. The destruction of natural resource is primarily the outcome of weak governance, inefficient bureaucracy, conventional industrial solutions, and inadequate financial planning (Welsch, 2008). Many studies have found a link between abundant use of natural resources and political upheaval. Furthermore, results revealed policies are ineffective to protect environmental deterioration (Ali, 2009; Alstine and Neumayer, 2010).

2.2. Nexus between FDI and CO_2 emissions

Since 1990s, after industrial revolution, it is believed that FDI intensifies the import of technology, increase economic activity, upgrade the infrastructure, speed up industrial growth, enhance financial activity, mobilize the production system which become the reason of further emissions specially in emerging and developing economies (Mahalik et al., 2021). Generally, FDI is inflow of funds from developed countries to developing countries, where they invest in assets and improve business operation with better technology, efficient and effective management expertise ultimately enhance business activity. With the flow of all these expertise foreign investors shows their keen interest to upgrade the industry in host country (Qamri et al., 2022). Over time this enhancement, this industrial enrichment is on the cost of environmental degradation and utilization of natural resources; governance system must design the green economy supportive policy mechanism which consider economic growth and environment simultaneously (Kulin and Johansson Sevä, 2019). FDI is to be considered one of the best ways for the transformation of expertise, technology, investments, infrastructure developments, new industrial paradigm, financial inventions, and advanced institutional policy developments from developed countries to emerging economies (Xu et al., 2021).

Although it discussed extensively in literature but still the dimensions of FDI with CO_2 are inconclusive (Shahbaz et al., 2018). This inconclusiveness exists because of three different theoretical dimensions. Firstly, the pollution-heaven hypothesis, which claims that weak institutional quality and environmental laws of host country may attract more FDI from profit intensive organization of developed countries to escape and evade from the costly regulatory polices compliance of the home countries (Shao et al., 2019). This will uplift the economic activity in host country at the cost of resources depletion and environment deterioration. Second, pollution halo dimension argues, multinational companies comply international standards concerning environmental protection rules and shift technological intensive infrastructure to support green economic development (Ali et al., 2020). This infers that investment inflows uplift the economic activity, advanced industrial evolution, and boost financing activities without deteriorating the environment (Naz et al., 2019). Finally, scale-effect dimension states that FDI inflows considerably contribute towards the host countries' industrial output which increases the consumption of fossil fuels energy leading to more utilization of NR and pollution (Pao and Tsai, 2011). Therefore, FDI needs strong financial system to manage financial needs of the business.

2.3. Nexus between financial development and CO_2 emissions

Financial development can directly influence economic growth of any economy, in many ways like ease of financing, availability of credit to new investment, financing to support domestic infrastructure, ease of channelizing funds and many more. Generally, financial development spurs your economic activity and promotes industrialization which ultimately increases pollution. A growing body of literature studied on financial development but due to different analysis techniques, methodology, sample selection (single or panel) country, financial development measures, environmental measures (CO_2 or footprints) produced different contradictory results. Over the past few years, the relationship amongst economic growth, FDI and financial development with economic energy get considerable attention and many researchers tested their interaction in different perspective. But literature has least discussed this variable with environment and given contradictory opinion on its dimension (Hao et al., 2020). Thus, we need more empirical studies to conclude the opinion (Acheampong et al., 2020; Saud et al., 2019). Some studies even pointed out the fact that financial development damages environmental quality by increasing FDI, attracting profit driven companies, enhance industry explosions, promote fuel-based industry solution and economic activity (Tamazian and Rao, 2010). Shahbaz et al. (2017) stated that financial development attracts more FDI that enhances energy consumption, however, when financial development collaborates with technology, it reduces emissions. Financial development decreases carbon emissions by financing energy effective investment, replaces traditional energy to renewable energy, minimizes energy consumption with automotive and modernized infrastructure (Hunjra et al., 2022).

2.4. Nexus between technology and CO_2 emissions

Technology innovation is considered as one of the finest tools to minimize carbon emissions by eliminating fossil fuel industrial solution, enhance advanced and automotive infrastructure, increase efficient use of energy, replace the energy intensive processes with information services, save natural resources, improve economic growth with sustainable and ecological business practices (Ahmad et al., 2021). The effects of ICT on environment are relatively complicated and literature gives inconclusive opinion on it. Theoretically, it discussed ICT impacts environmental quality through 3 modes. First, through the Price effects, when IT based solutions upsurge the demand of other energy intensive products due to decrease in their prices, the price conscious phenomena prevail more longer in developing countries and become the reason of more emission (Shabani and Shahnazi, 2019). The second mode is substitution which is minimizing the emission by offering restructuring,

IT based solution, energy efficient services, automotive production lines, especially decarbonization (Ozcan and Apergis, 2018), dematerialization, demobilization, replacement of physical products with virtual like books, manual works into email and software (Park et al., 2018), physical monitoring of transportation with management information system (MIS), GPS and AI (Shabani and Shahnazi, 2019). Advancement of ICT provide stewardship solutions, monitoring application, dematerialization options, advanced controlled and administrative solutions, web disseminative controlling, recycling, demobilization, and transport replacement jointly these all increase efficient use of energy and decrease emission (Baloch et al., 2022). The third mode stated that the innovation will increase production output, generate demand of new product, enhance the production unit and processes, more IT based devices produced ultimately increase energy consumption and affect negatively (Shabani and Shahnazi, 2019).

2.5. Nexus between governance and CO_2 emissions

Many theories have defined the relationship between governance and environment quality (Güney, 2022; Khan et al., 2022). The World Governance Indicators (WGI) defined it as: "Governance is set of traditions, rituals, laws, institutions and principles through which authority is exercise". It is composed with the set of rules, customs, rituals, and process through which an individual, organization, society or country governed and controlled. It is comprised of the process through which governments are selected, administered, controlled, and replaced; the capacity through which government formulate and implement polices, the way government run and controlled economic indicators by exercising powers through institutions to ensure equity, transparency and to empower regulators (Rafei, M. et al., 2022). Nevertheless, governments are still seeking answers to advance their governance model to ensure sustainable development, environment friendly economic growth, sustainable use of NR, and adaptive innovations. Good governance empowers the institution to plan and formulate the strategies required to achieve the SDGs. It includes many stakeholders at different levels like governments, individuals, organizations, civil society in decision making process to promote rule of law, autonomy, equality, transparency, inclusive and efficient management decisions so to enhance equitable distribution of resources for sustainable development.

3. Data and empirical Modelling

3.1. The data

The study used the longest available data for 19 Asian countries for the period 1996–2020 from the most reliable source, World Development Indicators. (CD-ROM, 2022). We used most likely measure carbon emissions (CO_2) metric tons per capita for environmental degradation and many researchers like (Baloch et al., 2019; Cheng et al., 2019) have used it in their studies. Governance (GOV) is measured by 6 WGI indicators: government effectiveness, political stability, rule of law, voice and accountability, regulatory quality, and corruption; these variables are measured on the basis of percentile rank and data obtained from WDI's database. FDI is measured as net flows (% of GDP) and technology innovation (TCI) by the total patents which are mainly used in previous studies (Alam and Murad, 2020). A measure of financial development is a proxy of credit to the private sector adopted by Levine (2005). Natural resources are measured as total natural resource's rents (%GDP). Gross domestic product (GDP) represents economic growth and is expressed as GDP per capita. Table-1 lists all study constructs along with their symbols, definitions and data sources used for the present study.

Table-1
Definition of Variable(s).

Variable	Codes	Description	Database
Carbon dioxide emission	CO_2	Carbon emissions (CO_2) per capita	WDI
Control of corruption	CC	It determines the degree of general perception in society for private gain. It considers all small and large sources of private income. If it is effectively controlled, it means that the regulatory body is powerful and dominant in policy implications.	WDI
Government effectiveness	GE	It measures the role of the government in assigning authorities with respect to the government's credibility, freedom, civil rights, public services, and commitment to all policies defined by the Constitution.	WDI
Political stability and lack of violence	PV	It determines the possibility government can be destabilize through the unconstitutional means. Determine the probability regime change with unconventional ways.	WDI
Regulatory quality	RQ	It assesses how effective the government is in terms of the rule of law and the implications of policies that promote private development and open markets.	WDI
Rule of law	RL	It describes the perception agents are confident in the application of laws, rules of society, safety measures, justice of courts, etc.	WDI
Voice and accountability	VA	It considered speech rights for citizens, how much freedom people had to express themselves, participate in general voting and to become part of government.	WDI
Foreign direct investment	FDI	Foreign direct investment, net inflows (% of GDP)	WDI
Technology	TCI	Number of total patent applications	WDI
Financial development	FD	Domestic credit to private sector (% of GDP)	WDI
Economic growth	GDP	GDP Per capita GDP (Constant 2010 US $)	WDI
Natural resources	NR	Total natural resources rents (% of GDP)	WDI

Note: Governance variables definition taken from World Bank and this measure refers to Halkos and Tzeremes (2013).

3.2. Empirical modeling

Our empirical model suggests that fossil fuel-based energy consumption, over exploitation of natural resources, weak institutional quality, fuel intensive FDI, less environment friendly fiscal policies and traditional consumption patterns would exert negative environmental externalities in economies. Currently, these countries ineffective stewardship policy cause for more natural resources depletion and mounting economic growth by attracting fuel intensive FDI. The high industrialization increased the use of natural resources and emit more emission. A weak stewardship policy of natural resources promote industrialization and production in developing countries. While, apparently, this trend benefits the host country's economy, such FDI inflows also increase pollution from unregulated industrial undertakings. Many businesses are participating in this global dispersal of investment, and many countries are increasing their use of FDI to accelerate their economic growth. Still, environmental glitches and resource depletion concealed behind such circumstances must not be disregarded. Sustainable use of resources and ecological development is the biggest challenge in current era. The countries claimed best governance system are overexploiting the resources and emitting more carbon. The challenge for governance is the need to design policy mechanisms that are systematics, fair, adaptive to climate change, assure sustainable use of resources, attract high-tech FDI and support green financing. In Asia, environmental governance, as in other parts of the world, has transformed considerably over the past few decades. The quest of boundless economic growth has been

prevailing for decades, still the region development track has been accustomed to the market demands and civil society for further growth at the cost of environmental degradation. With persistent reckless developments, demand for resources has also been mounted to a great extent. In 2020, economic growth has been slowed down due to Covid-19 crisis, but the region is expected to witness rapid growth in coming years, especially in heavily populated countries such as Pakistan, Indonesia, India, China, and Vietnam (OECD, 2020). If effective environmental mechanisms are not established in due time, series of local and global effects might be seen in the form of enormous demand of natural resources. Regardless of the incredible progressions in environmental governance in Asian region during the last decades, still countries have to take a long way to go so to cope with all range of environmental concerns.

In order to achieve high economic growth at an early stage of development, natural resources have been indecisively destroyed with little concern due to scale effects, leading to environmental quality degradation. Owing to composition effects, as the level of development increases, policymakers comprehend the need to curtail pollution and upgrade environmental quality through improved institutional quality, financial innovation, technological advancement, enforcement of steward policy for resources, environmental governance, and eco-friendly FDI. Improved governance is indispensable for environmental quality protection and ecological use of resources. It refers to sustainable natural resources and environmental conservation.

For this study, related variables were selected based on earlier empirical and theoretical foundations. CO_2 emission is considered the determining factor in the cause of environmental degradation and climate change. The ease of access to financial resources increases people's affordability and changes their consumption patterns. It also increases the production capacities of industry and the overall consumption of the economy which causes more emission. If financial policies are not eco-friendly, then its development has a negative impact on the environment. Financial development is measured by the proxy of domestic credit to private sector. Technology developments automate the industrial solution and reduces emission over time but environmental quality may also decline if there is no emphasis on providing green innovations and eco-friendly solutions. The abundance of natural resources increases productivity and is a source of competitive advantage and contributes to growth and economic power in modern era. It is measured by total natural resources rents. Finally, economic growth the most widely discussed association with CO_2 in literature is measured with GDP. It has an embryonic relation with the environment, initially having a negative impact on the environment but with later development and adoption of advanced processes, its impact diminishes with higher levels of economic development. Considering the above backdrop, the study composed an empirical model with the following equation(s).

$$(CO_2)_{it} = \alpha_0 + a_1(GOV)_{it} + a_2(FDI)_{it} + a_3(TECH)_{it} + a_4(FD)_{it} + a_5(GDP)_{it} + a_6(NR)_{it} + u_{i,t}\ldots\ldots \tag{1}$$

To capture non-linear association and to deal with exponential variation, we took the natural log of the variables. Shahbaz et al. (2012) recommends the natural-log of equation-1 to produce more robust and consistent results in panel data. So, the transformed equation-1 for the current model is given as following:

$$ln\ (CO_2)_{it} = \alpha_0 + a_1 ln(GOV)_{it} + a_2\ ln\ (FDI)_{it} + a_3\ ln\ (TECH)_{it} + a_4\ ln\ (FD)_{it} + a_5\ ln\ (GDP)_{it} + a_6\ ln\ (NR)_{it} + u_{i,t}\ldots\ldots \tag{2}$$

In above equation-2 CO_2 shows carbon emissions, Gov and FDI are representing governance and foreign direct investment respectively. However, FD, GDP and NR indicated financial development, economic growth and natural resources. Moreover, $u_{i,t}$ is showing error time with i to capture cross sectional aspect across country and time t..

4. Methodological Strategy

The empirical analysis of this study is performed in seven steps. First, the descriptive characteristics of the data are analyzed. Second, three different tests were applied to check cross-sectional dependence between panels. Third, we used a panel unit root test to identify order integration among variables. Fourth, Westerlund cointegration test is applied to test the cointegrating relationship. Fifth, the PMG estimator is used to estimate the error correction term. Sixth, dynamic threshold regression is employed to examine the threshold effects of governance on the relationships between FDI, FD, GDP, TCI, NR and CO_2 emissions. Finally, the iterated Generalized Least Square (IGLS) is used to check the robustness of the results. This technique is capable to deal with the issues of cross dependence, panel-wise heteroscedasticity, and autocorrelation (Alonso et al., 2017; Ikpesu et al., 2019).

4.1. Cross-sectional dependence tests

In the first step, we look for signs of cross-sectional dependency or time-dependent behavior in the data set. Countries are likely to have similar characteristics in terms of geo-graphics, cultural and economic aspects there is a probability the data may carry cross-sectional dependence and lead to inefficient results (Tugcu, 2018). We used Pesaran (2004) CD test, Friedman (1937) test, and Frees (2004) to address the issue of cross-sectional dependence. The null hypothesis of these tests states that there is no cross-sectional dependence. Pesaran's CD test statistic is estimated as follows:

$$CD = \sqrt{2T/(N(N-1)}\left(\sum_{i=1}^{N-1}\sum_{j=i+1}^{N}\widehat{\rho}_{ij}\right) \tag{3}$$

The CD test statistic are normally distributed with $N(0,1)$ for $N\rightarrow\infty$ and large T. The mean of this statistic is zero for fixed time and cross-sectional units. This test is reliable for panel data models having the characteristics of homogenous, heterogenous, dynamic, and non-stationary. Friedman (1937) test is a nonparametric test and works based on Spearman's rank correlation coefficient. If $\{r_{i,1},\ldots\ldots,r_{i,T}\}$ is the rank of $\{u_{i,1},\ldots\ldots,u_{i,T}\}$ and average rank equals $(T+1/2)$, the correlation coefficient is given below:

$$r_{ij} = r_{ji} = \frac{\sum_{t=1}^{T}\left\{r_{i,t} - \left(T+\frac{1}{2}\right)\right\}\left\{r_{j,t} - \left(T+\frac{1}{2}\right)\right\}}{\sum_{t=1}^{T}\left\{r_{i,t} - \left(T+\frac{1}{2}\right)\right\}^2} \tag{4}$$

The Friedman's test statistic is given as:

$$R_{ave} = \frac{2}{N(N-1)}\sum_{I=1}^{N-1}\sum_{J=I+1}^{N}\widehat{r}_{ij} \tag{5}$$

The Frees' test statistic is given below:

$$R^2_{ave} = \frac{2}{N(N-1)}\sum_{i=1}^{N-1}\sum_{j=i+1}^{N}\widehat{r}^2_{ij} \tag{6}$$

If $R^2_{ave} > (T-1)^{-1} + \frac{Q_q}{N}$, the null hypothesis is rejected. Q_q represents the quantile of the Q distribution.

4.2. Panel unit root tests

The CIPS test (Im et al., 2003) and the Cross-Sectionally Augmented Dicky Fuller (CADF) test (Pesaran, 2007) are used to test the unit root of the modeled variable(s). These second-generation tests were used effectively by many earlier studies to eliminate the cross-sectional dependence (Kang et al., 2021, Rahman and Sultana 2022). Pesaran (2007) used the following dynamic heterogenous model in panel data

with T-time and N cross-sectional elements.

$$Y_{i,t} = (1-\delta_i)\mu_i + \delta_i Y_{i,t-1} + u_{i,t} \tag{7}$$

where, $i = 1, \ldots.., N$ and $t = 1, \ldots., T$.

Cross-sectionally augmented form of the test is given below:

$$CIPS = \frac{1}{N}\sum_{i=1}^{N} CADF_i \tag{8}$$

4.3. Cointegration test

We proceed further with the Westerlund panel cointegration test (Westerlund, 2007) to check the long run cointegrating relationship between the dependent and explanatory variables. This test is capable to account for the cross-sectional dependence and heterogeneity between panels. In panel tests, Westerlund cointegration tests are used to compare the null hypothesis of no cointegration for the entire group of interdependent countries. The null hypothesis of this test is "no long run cointegration". The parameters of Westerlund cointegration test are as follows:

$$\Delta z_{it} = \sigma_i d_i + \theta_i \left(z_{i(t-1)} + \pi'_i y_{i(t-1)} + \sum_{j=1}^{m} \theta_{i,j} \Delta z_{i(t-j)} + \sum_{j=0}^{m} \varphi_{ij} \Delta y_{i(t-j)} + \omega_{it} \right. \tag{9}$$

Where Δz_{it} is generated with both constant and trend. $\Delta y_{i(t-j)}$ is independent of ω_{it} and furthermore, we assume error terms are independent across both i and t..

4.4. Estimation techniques and robustness

The standard panel data models provide inefficient results with the presence of cross-sectional dependance in data. To deal with this issue, this study applied a dynamic panel ARDL approach using the PMG estimator (Pisaran et al. 1999) to find the long-run coefficients. Unlike the Mean Group (MG) estimator, PMG provides stable coefficients in the presence of outliers (Adams et al., 2016). Several studies have used PMG to find long-run coefficients by incorporating cross-sectional dependence and heterogeneity between panels (Hashmi et al., 2022; Zaman et al., 2020). This study also uses a dynamic panel threshold regression model (Kremer et al., 2013) to examine the non-linear effects of FDI, FD, GDP, TCI, and NR on CO_2 emissions under different levels of governance. This technique is an extended version of Hansen's (1999) static panel threshold regression model. Because the present study analyzes the threshold effects of governance while controlling for GDP, this may introduce endogeneity bias and the biased results of governance threshold. Finally, robustness is checked using iterated panel GLS. The rationale behind choosing this technique lies in the fact that this method can address the issues of group-wise heteroscedasticity, first-order autocorrelation, and the cross-sectional dependence among panels (Cameron and Trivedi, 2009; Le and Nguyen, 2019).

5. Results and their discussion

A summary of descriptive statistics is shown in Table 2. The mean, standard deviation, minimum, and maximum values indicate that the data is smooth. Three different tests were used to examine cross-sectional dependence and their results are presented in Table 3. The probability value of CD test is 0.0006 which indicates that the null hypothesis of no cross-sectional dependence is rejected at the 99% confidence interval. Similar results are found in the Friedman's test. Likewise, the test statistic of Frees test also rejects the null hypothesis. Hence, the existence of cross-sectional dependence in the data has been proved.

In current scenario, it is important to know first the level of stationarity for empirical estimation in panel data. After the confirmation of cross-sectional dependence in the data, we applied second generation unit root test to check the order of integration among variables. The CIPS test is employed on three types of equations. First, without constant and trend, second, with only constant, and third with both constant and trend unit root tests are applied. These different specifications are used to obtain the robust results. GDP per capita is non-stationary across all three specifications. It is also non-stationary without constants and trends in the first difference equation, and also with constants. This variable is found to be stationary at the first difference in the equation with both constant and trend. Hence, GDP per capita is found to be integrated at order 1, I (1). Similarly, all other variables are also I (1). The detailed results of CIPS test are provided in Table-4. The CADF test is applied using two types of specifications i.e. one with constant, and the other with a constant and trend. The results indicate that all the variables are I (1). Here below, CADF results are presented in Table-5.

We need to figure out the cointegration first to get more reliable and efficient estimates. A panel cointegration method is required to solve the cross-sectional variability. In some cases, it is necessary to use it when the series elements of each cross-time sectional are shorter than those of the other cross-time sections. This study employed the Westerlund panel cointegration test by testing two alternative hypotheses. One is that some panels are cointegrated, and the other is that all panels are cointegrated. The p-values indicate that there is a long run cointegration relationship exists in the model. Here below Table-6 presents the test results.

We employed the PMG estimator to obtain the long-run coefficients and the error correction term. CO_2 emission (Dependent variable) is measured in metric tons per capita. The results are reported in Table-7. We have estimated three different equations by adding and dropping some explanatory variables to avoid the issue of collinearity among the variables and obtain the robust estimates. In all three equations, the error correction terms are −0.240, −0.277, and −0.262, respectively. This implies that deviations from long run equilibrium can be automatically corrected (Pahlavani and Rahimi, 2009). The coefficient of GDP per capita is found to be significant and positive, whereas its squared term is significantly negative. These findings confirm the Environmental Kuznets Curve (EKC) hypothesis of an inverted-U-shaped relationship between the level of development and pollution. It is inferred that initially, CO2 emissions increase with increasing income levels for selected Asian countries, however after attaining a certain level of income, CO2 emissions fall due to further increases in income. These findings are in line with earlier research findings (Apergis and Ozturk, 2015; Hashmi et al., 2022; Murshed et al., 2021). In equations (01) and (02), coefficients of governance index are reported. These

Table-2
Descriptive summary.

Variable	Mean	Std. Dev.	Min	Max
CO2	4.84	4.12	0.14	17.69
GDP	8.51	1.33	6.37	11.02
GOVINDEX	0.89	1.23	−1.80	2.64
TCI	7.69	2.81	1.09	14.24
FDI	6.97	24.91	−37.17	280.13
FD	71.26	54.92	3.01	255.31
NR	7.03	11.47	0.0002	55.47

Table-3
Cross-sectional dependence analysis.

Test	Test statistic	P-value/Critical value	Conclusion
Pesaran CD test	3.42	0.0006***	Cross-sectional dependence exists
Friedman's test	38.41	0.0034***	Cross-sectional dependence exists
Frees' test	3.10	0.1947***	Cross-sectional dependence exists

Note: *** denotes significance at 99%, and ** at 95% level.

Table-4
CIPS unit root analysis.

Variable	CIPS Statistic						Decision
	I (0)			I (1)			
	With no constant and trend	With constant	With constant and trend	With no constant and trend	With constant	With constant and trend	
GDPC	−0.947	−1.779	−1.443	−2.571	−2.313	−3.589***	I (1)
GOVINDEX	−0.372	−1.720	−2.478	−4.368***	−4.400***	−4.611***	I (1)
CO2	−1.841	−1.996	−1.875	−3.842***	−3.869***	−4.202***	I (1)
FD	−1.313	−1.242	−1.494	−3.756***	−3.744***	−4.395***	I (1)
FDI	−1.565	−2.521	−2.178	−5.219***	−5.203***	−5.196***	I (1)
TCI	−1.901	−1.760	−1.067	−4.785***	−4.855***	−4.869***	I (1)
NR	−2.000	−2.143	−2.846	−4.428***	−4.492***	−4.493***	I (1)

Note: *** denotes significance at 99%, and ** denotes significance at 95% confidence interval.

Table-5
CADF unit root analysis.

Variable	CADF P-Value				Decision
	I (0)		I (1)		
	With constant	With constant and trend	With constant	With constant and trend	
GDPC	0.293	0.991	0.002***	0.029**	I (1)
GOVINDEX	0.989	0.592	0.086*	0.000***	I (1)
CO2	0.253	0.977	0.001***	0.027**	I (1)
FD	0.721	0.651	0.000***	0.000***	I (1)
FDI	0.391	0.582	0.000***	0.000***	I (1)
TCI	0.113	0.689	0.000***	0.000***	I (1)
NR	0.996	0.840	0.000***	0.000***	I (1)

coefficients indicate that there is a long-run relationship between governance and CO2 emissions i.e. increase in governance index escalates CO_2 emissions in the selected Asian countries. Khan et al. (2022) also found similar results. However, after including squared term of the governance index in equation-3, it is found that governance has a non-linear association with CO_2 emissions. There can be several explanations of such a relationship. First, institutional governance promotes economic growth, which has a non-linear relationship with CO_2 emissions (Khan et al., 2022; Siddique et al., 2016). Second, institutional governance initially focuses on achieving high income levels but after a certain level is reached, further improvements in governance focus on environmental concerns. Technological innovation also has a positive long-term relationship with CO_2 emissions. These results suggest that technological innovation has not yet had a significant impact on the energy transition and carbon neutrality. Also, it is likely that the current level of technological innovations in selected Asian countries does not possess green innovations (Ulucak, 2021). FDI inflows also have a non-linear effect on CO_2 emissions. Unless FDI inflows reach a certain value, an increase in inflows leads to an increase in CO_2 emissions. Huynh and Hoang (2019) also reported similar results. The utilization of natural resources promotes economic growth at the cost of environmental degradation. A non-linear relationship with CO2 is observed, after attaining a certain level of development, production and consumption patterns reduce resource depletion. The increase in income level drive the economic growth, advance the infrastructure, adaptation of automotive and high-tech industry solutions reduces the consumption of natural resources. The overutilization of natural resources affects environment negatively and cause for more emissions. Adaptation of technology, sustainable use, automotive and advanced industrial solution, stewardship policy and renewable energy consumption can preserve the resources and clean environment. Financial development initially pollutes the environment, however after reaching a certain peak, further rise in FD improve environmental quality by reducing CO_2 emissions. This means that in the early stages of FD, the financial system of selected countries supports fossil fuel-based growth, while after reaching a certain point green financing becomes significant to reduce CO2 emissions. In summary, GDP, governance index, TCI, NR, and FD have long-run EKC type relationship with CO_2 emissions.

This study further performed a dynamic panel threshold regression model to study the asymmetric impacts of GDP, TCI, FDI, FD and NR on CO_2 emissions subject to the different levels of governance index of 19 selected Asian countries. The results are presented in Table-8. Put differently, CO2 emissions are the dependent variable, and the threshold variable is the governance index. Equation (01) provides the effect of technological innovation on CO2 emissions before and after the threshold level of governance. Here, the threshold value of governance index is 0.359. Before this level of the governance, an increase in technological innovation leads to an increase in CO_2 emissions, after this leads to decrease in CO_2 emissions. These findings indicate that surpassing the threshold level of governance is essential to promote environmental quality. The equation-2 explores the role of FDI on CO_2 emissions before and after a certain threshold value of governance index. The threshold level is 0.109. The results show that, an increase in FDI increases CO_2 emissions before the governance level of 0.109, and leads to decrease in CO_2 emissions after this governance threshold.

Table-6
Westerlund cointegration analysis.

Alternate Hypothesis	Test statistic	P-value
Some panels	2.0183	0.002***
All panels	2.6721	0.002***

Table-7
PMG long run analysis.

VARIABLES	1	2	3
	CO2	CO2	CO2
Error Correction	−0.175***	−0.254***	−0.156***
	(0.0523)	(0.0929)	(0.0452)
GDP	0.00173***	0.00106***	0.00101***
	(0.000138)	(0.000125)	(0.000116)
GDP^2	−3.45e-09***	−1.93e-08***	−4.65e-09**
	(1.33e-09)	(8.00e-10)	(2.04e-09)
GOVINDEX	0.568***	0.481***	0.460*
	(0.179)	(0.0965)	(0.256)
$GOVINDEX^2$	–	–	−0.159***
			(0.0273)
TCI	0.247***	0.412***	0.383***
	(0.0626)	(0.0616)	(0.0769)
FDI	0.0213***	–	0.0600***
	(0.00711)		(0.00540)
FDI^2	−0.000184**	–	–
	(9.40e-05)		
NR	0.0719***	0.0263***	0.106***
	(0.0170)	(0.00314)	(0.0237)
FD	–	0.00553*	0.00891***
		(0.00321)	(0.00339)
FD^2	–	−3.07e-04***	–
		(1.27e-05)	
Observations	456	456	456

Table-8
Dynamic Threshold Model (Threshold Variable is GOVINDEX).

VARIABLES	1	2	3	4	5
	CO2	CO2	CO2	CO2	CO2
Lag CO2 Before	1.160***	0.960***	−0.0678	0.860***	0.744***
	(0.0361)	(0.00336)	(0.494)	(0.109)	(0.0217)
Lag CO2 After	−0.157***	−0.0832***	1.197**	−0.163**	0.0170
	(0.0365)	(0.00478)	(0.465)	(0.0790)	(0.0834)
Threshold	0.359***	0.109***	0.517***	0.392***	0.359***
	(0.0155)	(0.0139)	(0.0981)	(0.0764)	(0.0364)
GDP Before	0.000350***	–	–	–	–
	(4.92e-05)				
GDP After	−0.000328***	–	–	–	–
	(5.13e-05)				
TCI Before	–	1.94e-07***	–	–	–
		(1.33e-08)			
TCI After	–	−1.58e-06***	–	–	–
		(2.03e-08)			
FDI Before	–	–	0.138*	–	–
			(0.0835)		
FDI After	–	–	−0.136*	–	–
			(0.0822)		
FD Before	–	–	–	0.0104**	
				(0.00438)	
FD After	–	–	–	−0.00791	
				(0.00557)	
NR Before	–	–	–	–	0.00754***
					(0.00096)
NR After	–	–	–	–	−0.0914***
					(0.00260)
Constant	−3.186***	0.964***	−8.612**	−1.256**	3.333*
	(0.477)	(0.0440)	(3.781)	(0.624)	(1.991)

Similarly, the results of equation (05) show that before the governance level of 0.359, the use of natural resources increases carbon emissions and after that level decreases emission due to growth and development in infrastructure. These results are consistent with the findings of Huynh and Hoang (2019). There are similar asymmetric effects of GDP on CO2 emissions but with different governance threshold values.

This study also performed iterated GLS method to confirm the robustness of the empirical results. In order to draw valid inference, the robustness of the experimental results should be checked in two ways. First, whether the estimated coefficients are robust to changes in the number of explanatory variables and second, competing with estimation techniques. The first type of robustness is performed in the main empirical analysis and the second is confirmed by GLS. Robustness check are necessary where the coefficient of core variables is insensitive of inclusion or omission of variable. This irrelevance of coefficients is conclusive to draw casual inference between variables under appropriate conditions. The robustness check is also strengthened in terms of inclusion and exclusion of some explanatory variables. This method is assumed more valid because it directly considers heteroskedasticity (as well as cross-sectional and serial correlations in the estimation). The empirical results of GLS are reported in Table-9. The results show that the impacts of GDP, governance index, TCI, FDI, NR and FD are robust against the change in estimation technique and relevant explanatory variables.

Table-9
Iterated GLS Analysis (Dependent Variable is CO_2).

VARIABLES	1	2	3
	CO2	CO2	CO2
GDP	0.000508***	0.000508***	0.000516***
	(3.67e-09)	(3.40e-09)	(9.16e-08)
GDP^2	−5.46e-09***	−9.36e-09***	−8.21e-09***
	(3.16e-10)	(2.57e-10))	(1.58e-10)
GOVINDEX	0.0820***	0.103***	0.105***
	(1.45e-06)	(1.64e-06)	(3.54e-05)
$GOVINDEX^2$	0.0195***	0.0180***	0.0862***
	(8.71e-07)	(8.05e-07)	(2.26e-05)
TCI	0.00264***	0.00458***	0.0141***
	(6.08e-08)	(8.53e-08)	(3.73e-06)
FDI	–	−9.58e-06***	−5.07e-05***
		(3.03e-10)	(1.46e-08)
FDI^2	0.000407***	0.000417***	0.00350***
	(8.22e-08)	(6.59e-08)	(2.38e-06)
FD	–	–	−1.93e-05***
			(2.12e-08)
FD^2	0.000508***	0.000508***	0.000516***
	(3.67e-09)	(3.40e-09)	(9.16e-08)
NR	0.00896***	0.00879***	0.0193***
	(9.79e-08)	(7.44e-08)	(3.10e-06)
Constant	0.450***	0.441***	0.548***
	(5.38e-06)	(5.20e-06)	(0.000110)
Observations	475	475	475
Number of ID	19	19	19

6. Conclusion and policy implications

The main focus of the current study was to evaluate natural resources reserves importance in the context of COVID-19 and global conflicts, i.e. Russia-Ukraine war. The study is of great importance especially considering Asian economies which has abundance of natural resources. A nation with rich and diverse natural resources can expand more quickly than a state with fewer resources. The study further investigates either natural resources is a blessing or a curse for achieving sustainable development or cause to decline sustainable development. The purpose of study was to design effective and conflict-sensitive management solutions for natural resources and sustainable development. We used a series of econometric tests to check cross-sectional dependence, robustness, heterogeneity, and stationarity of the data. Three different CD tests such as (Friedman, 1937), Freeze test (Frees, 2004) and (Pesaran, 2004) are performed for cross-sectional dependence. These tests rejected the null hypothesis of cross-sectional dependence. In addition, the second-generation test of CIPS and CADF are used to deal with cross-sectional dependence issues. The CIPS test is based on three

types of equations. Furthermore, we applied (Pesaran, 2007) to check data stationarity and (Westerlund, 2007) to identify long-run cointegration between variables.

The coefficients are estimated by using PMG estimator, unlike the Mean Group (MG) estimator the PMG provides stable coefficients in presence of outliers. The coefficient of GDP per capita is found to be significant and positive, whereas its square terms are significantly negative. These findings confirm the EKC hypothesis of inverted-U shaped relationship between the level of development and pollution. It is inferred that initially, with the increase in income levels of the selected Asian countries, CO_2 emissions rise, however after attaining a certain level of income, CO_2 emissions fall because of further increase in income. There is long run association between governance and CO_2 emissions. A similar long run association observed between technology innovation and CO_2 emissions. FDI inflows have also non-linear impact on CO_2 emissions. Unless the FDI inflows reach a certain value, increase in the inflows leads to rise in CO_2 emissions. The trend of FD association with CO_2 is also observed similar to governance and technology innovation, initially FD causes for more emissions and pollutes environment but later on it improves environment quality with the increase of development level. The natural resources and CO2 emission trend is also similar to governance, TCI, and FD. In initial stage the consumption of natural resources deteriorates the environment but after a certain time with the advancement of infrastructure and development it started to decreases. All of these states have same like weak governance and institution, high level of social and economic inequality leads to ill equipped infrastructure to deal with climate change and resources depletion. This can exacerbate the tensions and fragility with the potential to lead violent conflict among countries. In summary, GDP, governance index, FDI, NR, TCI and FD have long run EKC type relationship with CO_2 emissions. This study also employs dynamic panel threshold regression model (Kremer et al., 2013) to examine the non-linear effects of FDI, FD, GDP, NR, and TCI on CO_2 emissions by subject to different levels of governance. These finding indicate that surpassing the threshold level of governance is essential to promote environmental quality and preservation of natural resources. The results show that, rise in FDI increases CO_2 emissions before this level of governance, and leads to fall in CO_2 emissions after this governance threshold. GDP, NR and FD also have the similar asymmetric impacts on CO_2 emissions but with different threshold values of governance.

6.1. Policy implications

Competition on natural resources including land, water, oil, forest – is a major role for a conflict specially in fragile states. Following are the suggestions for handlers.

1. A systematic, fair, inclusive, accountable, transparent, strong, and responsive natural resource governance system is essential to effective climate change adaptation and mitigation efforts and integral to control conflicts.
2. Good governance needed to avoids or mitigate the resource curse by implementing law and order, transparent and accountable supremacy systems, investing in social programs, and diversifying their economies.
3. The region needs to promote steward policy for resources. This can take the form of nationalizing resource assets, increasing taxes and royalties on resource extraction, and requiring foreign companies to work with local partners to ensure sustainable development.
4. The era has changed and demand for a fiscal policy supportive to renewable energy projects and support high-tech FDI. Only a few countries in the region like China and Japan have introduced green bonds.

Declaration of competing interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Data availability

Data will be made available on request.

Appendix

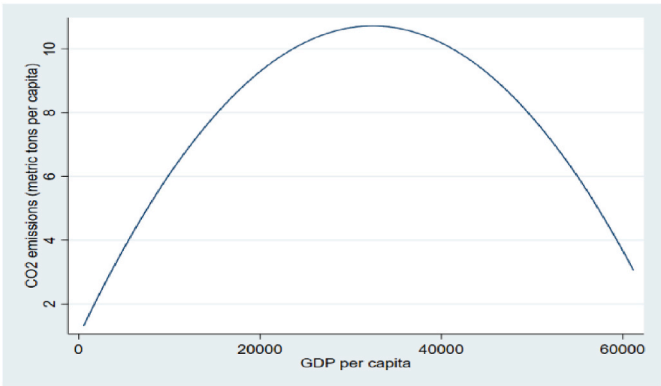

Fig. A1. GDP per capita and CO_2 emissions.

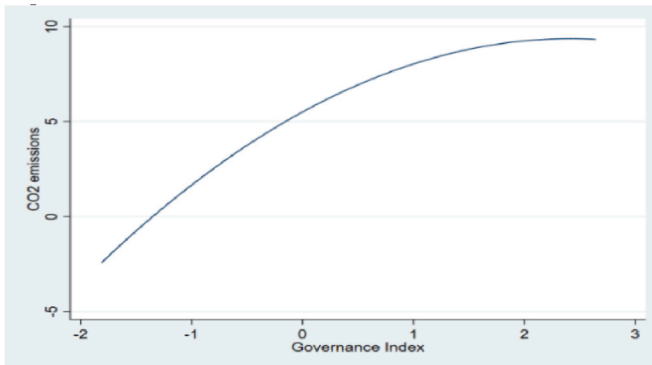

Fig. A2. Governance and CO_2 emissions.

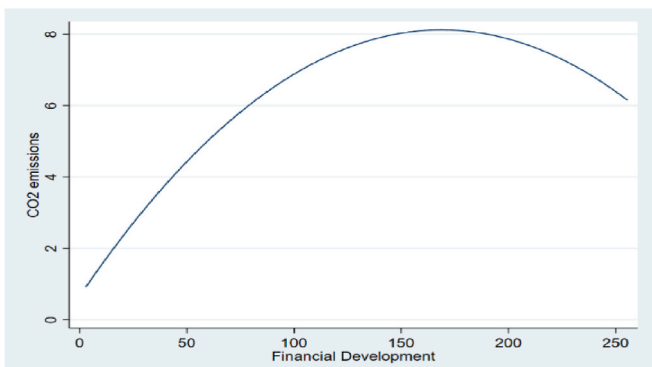

Fig. A3. FD and CO_2 emissions.

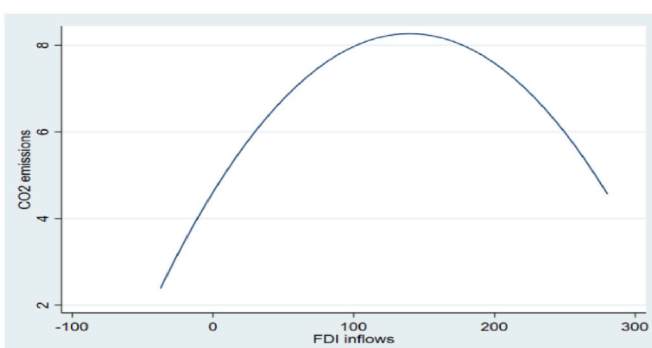

Fig. A4. FDI and CO_2 emissions.

Countries Excluded					
Sr. No	Name	Missing Variables	Sr. No	Name	Missing Variables
1	Afghanistan	TCI, FDI, FD	16	Maldives	TCI. FD
2	Armenia	FD, TCI	17	Myanmar	TCI
3	Azerbaijan	FD, TCI	18	Nepal	TCI
4	Bahrain	TCI	19	Oman	TCI
5	Bhutan	TCI	20	Qatar	TCI
6	Brunei Darussalam.	TCI, FD	21	Syria	TCI, FD, FDI
7	Cambodia	TCI	22	Tajikistan	TCI
8	Georgia	TCI	23	Thailand	FD
9	Indonesia	TCI	24	Timor-Leste	TCI
10	Iraq	TCI	25	Turkey	FD
11	Kazakhstan	FD	26	Turkmenistan	TCI, FD
12	Korea, Dem.	TCI, FDI, FD	27	United Arab Emirates	TCI, FD
13	Kuwait	TCI	28	Uzbekistan	FD
14	Lao PDR	TCI			
15	Lebanon	TCI			

- FDI: Foreign direct investment.
- TCI: Technology innovation.

- FD: Financial development.

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